Physics-Based Failure Analysis: Why Industry Models Will Default

Executive Summary

Using our proprietary dataset combining 777 days of controlled lysimeter data with 145,000+ tonnes of field deployment data, we conducted a rigorous physics-based analysis of the ERW industry's core assumptions. Our findings reveal three critical failure points that will cause the majority of existing ERW carbon credit models to default.

Failure Point 1: The Passivation Layer

What the Industry Assumes

Most ERW companies model basalt dissolution as a continuous, linear process over 5–10 years. Their financial models assume that a tonne of basalt applied today will steadily release ions and capture CO_2 for a decade.

What Our Data Proves

Basalt dissolution follows a strict front-loaded pattern:

- **Year 1:** Massive ion flush during initial rainfall events (the "Year 1 Flush")
- **Year 2+:** Complete reversion to background levels

The mechanism: As basalt weathers, it forms a silica-rich and iron-oxide residual layer on the grain surface. This "passivation layer" is physically impenetrable to water, permanently halting further dissolution. This is not a theoretical concern — it is documented in peer-reviewed geochemistry literature and confirmed by our continuous 777-day dataset.

Financial Impact

Any company selling 10-year carbon credit delivery contracts based on linear dissolution will face massive shortfalls beginning in Year 2. The credits simply will not materialize.

Failure Point 2: The Crusting Trap (Non-Linear Dose Response)

What the Industry Assumes

More rock = more carbon. If 100 t/ha captures X tonnes of CO_2, then 400 t/ha should capture 4X.

What Our Data Proves

The dose-response relationship is catastrophically non-linear:

Metric	Expected (Linear)	Observed
$^{400}/_{100}$ Ca Export Ratio	4.0×	−58.6×
400 t/ha vs Control	Strongly positive	Severely negative (−1,370,000 μmol)

At high application rates, the basalt physically clogs soil pores, reduces hydraulic conductivity by up to 27.5%, and reduces porosity by 6.3% (confirmed by independent global synthesis studies). The soil literally suffocates. Water cannot percolate through the rock layer, so the ions never flush into the leachate — and therefore never capture CO_2.

Financial Impact

Companies investing millions in high-dose applications and ultra-fine grinding equipment are destroying their own CAPEX. The optimal strategy is lower doses with coarser grain sizes deployed over larger areas.

Failure Point 3: The AI/ML Black-Box Trap

What the Industry Assumes

Deep Learning and machine learning models can accurately predict multi-year carbon drawdown from soil and weather variables.

What Our Data Proves

Non-linear ML models will:

1. Overfit to the Year 1 Flush and extrapolate it indefinitely

2. Fail to capture the physical mechanism of passivation (which is a binary threshold, not a smooth curve)
3. Be rejected by carbon registries that require deterministic, auditable MRV methodologies

The only defensible approach is white-box, physics-based modeling using established geochemical laws (e.g., Kohlrausch's Law for electrical conductivity relationships) calibrated against continuous sensor data.

The Correct Operational Model

Based on our analysis, the only profitable ERW deployment follows these constraints:

Parameter	Optimal Range	Rationale
Application Rate	15–25 t/ha	Avoids crusting threshold
Grain Size	Coarse crusher dust (d50 ~38 μm)	Maintains soil hydraulic conductivity
Target Soil	Low-pH, degraded (Q1 quartile)	Maximizes both yield gain (+47%) and Year 1 carbon flush
Logistics Radius	<$20/tonne delivered	Hard break-even limit at current carbon prices
Credit Horizon	Year 1 only	Only viable credit window before passivation
MRV Method	Deterministic physics (EC sensors + isotopic verification)	Only approach that passes registry audits

What We Are Not Sharing Here

- Our exact kinetic model parameters and calibration coefficients
- The specific geographic coordinates where these constraints have been validated
- Our proprietary logistics optimization algorithm
- The exact month/day when passivation permanently stalls dissolution